Twitter Sentiment Analysis

Apple & Google Products

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9,093
Labeled Tweets

3
Sentiment Classes

3
Models Trained

The Business Problem

Apple and Google receive thousands of Tweets/public opinions daily discussing their products and services on Social Media. Brand managers cannot manually monitor the thousands of tweets as is. The goal of this project is to build a machine learning model capable of automatically classifying tweet sentiment as positive, negative, or neutral.

📊

Volume

Thousands of tweets posted daily. No human team can keep up at this scale.

⚡

Speed

Negative sentiment goes viral within hours. Manual spotting is always too late.

🎯

Insight Gap

Without data, teams are guessing which products frustrate vs delight customers.

Our Solution: A machine learning model that reads and classifies tweet sentiment automatically — in milliseconds, at any scale.

The Dataset

9,093

Labeled Tweets

3

Sentiment Classes

~6%

Negative Tweets

Key Challenge

Negative tweets = only 6% of data.

A lazy model predicting 'No emotion' scores 60% accuracy while missing EVERY negative tweet.

This is why we use Macro Recall, not accuracy, as our primary metric.

Our Approach

A 5-step NLP pipeline from raw tweet to sentiment prediction

1

Load & Explore

9,093 tweets loaded. Inspected class balance, nulls, and brand distributions.

2

Clean Text

Removed URLs, @handles, punctuation. Applied stopwords removal and lemmatization.

3

TF-IDF

Converted words into 5,000 numeric features capturing their importance per tweet.

4

Train Models

Three models trained on 80% of data, tested on the held-out 20%.

5

Evaluate

Compared using Macro Recall — correctly penalises minority class misses.

Model Results

Primary metric: Macro Recall — averages recall equally across Positive, Negative and Neutral classes

Macro Recall is our primary metric. Below shows how each model performed on the hardest class — Negative emotion.

0.60

Negative Class Recall

Logistic Regression

0.48

Negative Class Recall

LinearSVC

0.21

Negative Class Recall

XGBoost

Full Confusion Matrix — Correct Classification Rate Per Class

Sentiment Class	Logistic Regression	LinearSVC	XGBoost
Negative emotion	**0.60**	0.48	0.21
Neutral / No emotion	0.98	0.97	0.98
Positive emotion	0.81	0.86	0.89

Why Logistic Regression Won

A surprising result — simpler models outperform trees on sparse text data

WINNER

Logistic Regression

- Negative Recall: 0.60 — highest
- Built for sparse TF-IDF features
- Direct word importance scores
- Fast, interpretable, deployable

2ND PLACE

LinearSVC

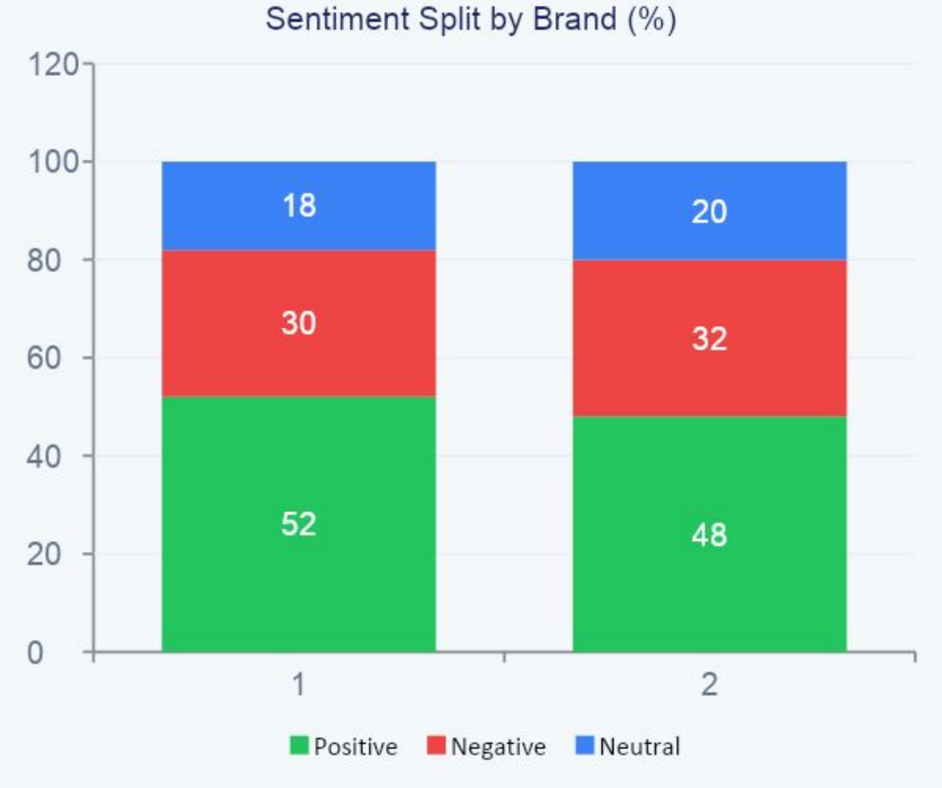
- Negative Recall dropped to 0.48
- Better Positive recall (0.86)
- Traded minority recall for margin
- Valid NLP model, wrong problem fit

NOT SUITABLE

XGBoost

- Negative Recall collapsed to 0.21
- 69% of negatives called Positive
- Trees can't split sparse TF-IDF
- Best for dense tabular data only

What the Model Found



Apple — Negative Drivers

- Battery life complaints
- Pricing too high
- Software crashes
- Repair costs

Google — Negative Drivers

- Privacy concerns
- Product discontinuations
- Ad frequency
- Support hard to reach

Both brands ~50% positive. The value is in the negative cluster — specific, actionable product feedback.

Recommendations for Stakeholders

>> 01 Automate Sentiment Monitoring

Right now, someone has to manually read tweets to understand how customers feel. Our model can do this automatically, reading thousands of tweets per minute and labelling each one as Positive, Negative, or Neutral — instantly.

>> 02 Get Alerts When Customers Are Unhappy

Whenever the model detects a Negative tweet, it can trigger an automatic alert to the relevant team. This means Apple or Google can respond to complaints within hours instead of days — before the problem goes viral.

>> 03 Track Sentiment Trends Week by Week

By running the model continuously, the team gets a weekly report showing whether sentiment is improving or declining. For example: did a new product launch make customers happier or trigger more complaints?

>> 04 Understand Which Products Need Attention

The model can be extended to filter by product — iPhone, MacBook, Pixel, Android — so product managers know exactly which product is generating the most negative feedback and where to focus their resources.

>> 05 Use Word Insights to Guide Marketing

The model reveals which specific words appear most in negative tweets — for example 'battery', 'price', 'crashing'. Marketing teams can use this to directly address customer pain points in their messaging and campaigns.

Key Takeaways

01

We built a model that classifies tweet sentiment with 0.60 Negative Class Recall — the metric that matters most for catching customer complaints early.

02

Logistic Regression beat XGBoost and LinearSVC because linear models outperform trees on sparse TF-IDF text features.

03

Apple and Google both enjoy ~50% positive sentiment. Negative feedback clusters around battery, pricing, privacy, and product discontinuations.

04

The model is production-ready. Deployed to a real-time pipeline, it classifies thousands of tweets per minute with no manual review.

Thank you